PassFrame
tuition

Statistics & Averages

Year 5 Accelerated — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: A pie chart shows 200 pupils' favourite drink: Water=45%, Juice=30%, Squash=25%. How many more pupils chose water than squash?

Method & answer: Water = 45% of 200 = 90. Squash = 25% of 200 = 50. Difference = 90 − 50 = 40.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the mean of 16, 24, 20, 28, 12.

2. Find the mean of 31, 37, 25.

3. Find the median of 18, 32, 11, 25, 19.

4. Find the median of 15, 33, 9, 39, 23.

5. Find the mode of 14, 19, 19, 25, 10, 19.

6. Find the mode of 23, 23, 30, 30, 30, 16.

7. Find the range of 26, 58, 40, 74, 44.

8. Find the range of 120, 90, 150, 80.

9. Which of these is the mode of 16, 16, 21, 28, 28, 28? A) 28 B) 16 C) 21 D) 22

10. A table shows pets: Dogs=46, Cats=31, Fish=22, Birds=25. How many pets are there in total?

11. A café's coffee sales over 4 days: Mon=74, Tue=101, Wed=58, Thu=87. What is the mean number sold per day?

12. Select ALL that apply. A frequency table shows: 1 pet = 16 pupils, 2 pets = 21 pupils, 3 pets = 10 pupils. Which statements are true? A) 47 pupils were surveyed in total B) The mode number of pets owned is 2 C) The mode number of pets owned is 1 D) More pupils own 2 pets than own 3 pets

13. Five friends scored 74, 80, 87, 76, 83 in a test. What is their mean score? A) 80 B) 81 C) 79 D) 82

14. Find the mean of 18, 18, 18, 18, 28.

15. Find the range of 38, 71, 53, 22, 59.

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the median of 23, 36, 18, 29.

2. Find the mean of 45, 57, 39, 67.

3. A library's book loans by class:

6. A bar chart shows sales at 4 shops: A=52, B=67, C=35, D=58. What is the mean sale per shop?

7. Find the mode of 13, 19, 19, 23, 27, 19.

8. A pets survey shows: Dogs=41,

11. Which of these is the range of 91, 46, 73, 109, 62? A) 63 B) 109 C) 46 D) 67

12. Find the median of 25, 13, 35, 19, 45, 29.

13. A pie chart shows a survey of 88

Amy=27, Ben=35, Cara=22, Dan=40. What is the total number of loans?

4. Which of these is the median of 22, 41, 30, 59, 15? A) 30 B) 41 C) 22 D) 59

5. Find the range of 56, 108, 82, 37, 95.

Cats=33, Fish=25, Birds=17. If 15 more fish are added, how many pets are there altogether now?

9. Select ALL that apply. A class recorded favourite subjects: Maths=36, Art=25, Science=19. Which statements are true? A) 80 pupils were asked in total B) Maths is the mode (most popular) C) Science is the mode (most popular) D) More pupils chose Art than Maths

10. Find the mean of 28, 40, 32, 46.

pupils: half prefer football, a quarter prefer swimming. How many prefer something else?

14. Which of these is the mean of 42, 50, 58, 66? A) 54 B) 56 C) 52 D) 58

15. Find the range of 149, 202, 171, 182.

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A shop's mean daily takings over 6 days is £84. What was the total takings over the 6 days?

2. The mode of a set of numbers is 15, and the range is 9. If the smallest number is 15, what is the largest number?

3. The mean of four numbers is 26. Three of the numbers are 18, 24 and 32. What is the fourth number?

4. A pie chart is split into just two colours so that purple is five times as big as orange. What is the angle for orange?

5. Seven numbers, in order, are 12, 18, 25, ?, 41, 48, 55, and their median is 33. What is the missing 4th number?

6. A team's points in 6 games were 50, 35, 45, 30, 55, 25. Find the mean and the range.

7. A survey of 150 pupils' travel is shown in a pie chart: 40% walk, 24% cycle, and the rest come by car. How many travel by car?

8. A set of 9 numbers has a mean of 34. If one of the numbers, 18, is removed, what is the mean of the remaining 8 numbers? A) 36 B) 34 C) 38 D) 32

9. A pie chart (360°) shows how 180 pupils travel to school: Walk = 200°, Car = 100°, Bus = 60°. How many pupils travel by car or bus combined?

10. Select ALL that apply. A class of 32 pupils' test scores have mean 71 and range 48. Which statements MUST be true? A) The total of all 32 scores is 2,272 B) At least one score is below 71 C) The highest score minus the lowest score is 48 D) Every pupil scored above 60

11. A trail-mix recipe (360° pie chart, total 1,080g) shows: nuts = 160°, raisins = 110°, seeds = 50°, and the rest is chocolate. What is the weight of chocolate, in grams?

12. The mean of 6 numbers is 38. Five of the numbers are 30, 34, 40, 44 and 36. What is the sixth number?

13. A bag of 10 numbers has a mean of 15. Which of these MUST be true? A) The total of the 10 numbers is 150 B) Every number in the bag is 15 C) The largest number is greater than 15 D) The smallest number is less than 10

14. Two classes of 30 pupils each recorded a mean of 1.8 pets per pupil in Class A and 1.2 pets per pupil in Class B. What is the mean number of pets per pupil across both classes combined?

15. A pie chart shows exam grades for 60 pupils: A* = 90°, A = 120°, B = 90°, C = 60°. How many more pupils got grade A than grade C?

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